

Matthew Egeler
CHE 312
HW 8
DUE 08/MAR/2020

78/100

9.15 1) Door $1 \times 1.8 \text{ m}$ $T_{\text{air}} = 15^\circ \text{C}$

A) Hot air rises so bottom is colder. boundary layer?

B) $T_s = 0^\circ \text{C}$ $\epsilon = .94$

$q_{\text{loss}} = q_{\text{conv}} + q_{\text{radiation}} \Rightarrow A [\bar{h} L (T_\infty - T_s) + (\epsilon)(T_{\text{sur}}^4 - T_s^4)]$
 $T_F = \frac{273 + 288}{2} = 280.5 \text{ K}$ Properties $V = 14.11 \times 10^{-6} \text{ m/s}$ $k = .0247 \text{ W/mK}$
 $\alpha = 1.986 \times 10^{-5} \text{ m}^2/\text{s}$ $Pr = .710$

$Nu_L = \frac{\bar{h} L}{k}$ $Nu_L = \left[.825 + \frac{.837 Ra_L^{1/4}}{\left(1 + (.492/Pr)^{1/4}\right)^{1/4}} \right]^2$

$Ra_L = \frac{g \beta (T_s - T_\infty) L^3}{\alpha V} = \frac{9.8 \cdot .0025 \cdot (15 - 0) \cdot 1.8^3}{1.986 \times 10^{-5} \cdot 14.11 \times 10^{-6}} = 7.65 \times 10^9$

$Nu_L = \left[.825 + \frac{.837 (7.65 \times 10^9)^{1/4}}{\left(1 + \left(\frac{.492}{.710}\right)^{1/4}\right)^{1/4}} \right]^2 \Rightarrow .825 + \frac{37.15}{(1 + .824)^{1/4}}$

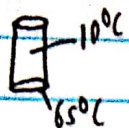
$= 31.97$

$\bar{h} = \frac{Nu_L k}{L} = \frac{31.97 \cdot .0247}{1.8} = .439 \text{ W/m}^2 \text{K}$

$q = 1.8 \cdot 2 \left[.439 (15 - 0) + .946 (288^4 - 273^4) \right]$
 $= 11.7 + 127.13 = 138.83 \text{ W}$

$\$ = .138 \text{ kW} \cdot 18 \text{ MW} = .025 \$ \cdot \left(\frac{24}{1 \text{ day}} \right) = \frac{.60 \$}{\text{day}}$

9.28 2)



0.5m

$$T_F = \frac{283 + 326}{2} = 304.5 \text{ K}$$

Properties: $\nu = 288 \times 10^{-6} \text{ m}^2/\text{s}$, $\mu = 1.45 \text{ W/mK}$

$$L = \frac{A_s}{A} \frac{\pi D^2/4}{\pi D} = \frac{0.014}{4} = 0.0035 \text{ m} \quad \alpha = 1.847 \times 10^{-7} \text{ m}^2/\text{s} \quad Pr = 3400$$

$$B = 0.7 \times 10^{-3}$$

$$q = \bar{h} A_s (T_s - T_\infty) \quad \bar{h} = \frac{Nu \mu}{L} \quad Nu =$$

$$Ra = \frac{g B (T_s - T_\infty) L^3}{\nu \alpha} = \frac{(9.8)(0.7 \times 10^{-3})(65 - 100)(0.0035)^3}{(288 \times 10^{-6})(1.847 \times 10^{-7})}$$

$$= 3.02 \times 10^7$$

$$\text{From 9.31 } \bar{Nu} = 0.45 Ra^{1/3} = (0.45) \sqrt[3]{3.02 \times 10^7} = 46.72$$

$$\bar{h} = \frac{Nu \mu}{L} = \frac{(46.72)(1.45)}{0.0035} = 54.19 \text{ W/m}^2\text{K}$$

$$q = (54.19) \pi (0.014)^2 (55) = 585.2 \text{ W}$$

9.20 3)

$$q = m c_p (T_{hi} - T_{ho}) = (0.1)(1400)(100 - 50) = 9500 \text{ W}$$

$$q = m c_p (T_{wo} - T_{wi}) = 0.1(1400)(T_{wo} - 30) = 9500$$

$$T_{wo} = \frac{9500 + 12600}{420} = 52.62^\circ\text{C}$$

(18/25)

$$\text{For counter flow Hex } q = UA \Delta T_{lm} \quad \Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln \Delta T_1 / \Delta T_2}$$

$$\Delta T_{lm} = \frac{(T_{ho} - T_{ci}) - (T_{hi} - T_{co})}{\ln \frac{(T_{ho} - T_{ci})}{(T_{hi} - T_{co})}} = \frac{(60 - 30) - (100 - 48.2)}{\ln \left(\frac{60 - 30}{100 - 48.2} \right)} = 40^\circ\text{C}$$

$$q = U (\pi D L) \Delta T_{lm} \quad L = q / (U \pi D \Delta T_{lm}) = 7600 / (0.025 \cdot 60 \cdot 40)$$

$$L = 40.3 \text{ m} \quad \times 69.3 \text{ m}$$

4) 11.26 A $\epsilon_1 = \dot{m}^{-.25} = \frac{q_1}{q_{max}}$ $\epsilon_2 = 2\dot{m}^{-.25} = \frac{q_2}{q_{max}}$

$$\frac{q_1 - q_2}{q_1} = \text{increase} = 1 - \frac{q_2}{q_1}$$

$\left(\frac{15}{25} \right)$

$$\frac{q_2}{q_1} = ? \quad \frac{\epsilon_1}{\epsilon_2} = \frac{q_1/q_{max}}{q_2/q_{max}} = \frac{q_1}{q_2} = \left(\frac{\dot{m}}{2\dot{m}} \right)^{-.25} = 1.19$$

$$\frac{q_2}{q_1} = \frac{q_1}{q_2}^{-1} = .841$$

$$(1 - .841) \times 100 = \boxed{15.91\%}$$

B) $\epsilon_1 = \dot{m}^{-.25} = \frac{q_1}{q_{max}}$ $\epsilon_2 = 2\dot{m}^{-.25} = \frac{q_2}{q_{max}}$ $\frac{\epsilon_1}{\epsilon_2} = \left(\frac{\dot{m}}{2\dot{m}} \right)^{-.25} = 1.07$

$$q = \dot{m} c_p \Delta T \quad \frac{\dot{m}_1 c_p (303 - 263)}{1.5 \dot{m}_2 c_p (T_2 - 263)} = 1.07$$

Using solver gives $T_2 = 287.922 K = \boxed{14.92^\circ C}$

X